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Physics-Guided Deep Learning Method for Tool Condition Monitoring in Smart Machining System

Shenshen Li, Xin Lin, Hu Shi, Yungao Shi, and Kunpeng Zhu

***Abstract*—Accurate and fast prediction of tool conditions is fundamental to improve the machining accuracy and consistency in smart machining systems. The current tool condition monitoring methods, i.e., physics-based and data-driven approaches, have either low prediction accuracy or model generalization. To solve the shortcomings and utilize the benefits of both sides, a novel physics-guided deep learning model is developed in this study. The physics of cutting mechanics and tool wear model is applied to guide the model construction and regulate the network learning process. The model first generates large labeled simulation dataset for the pretraining of the deep network, and solves the problem of labeled-data insufficiency for network training in practice. It then fine-tunes the model through the monitored signal to optimize the network weight and alleviate the deviation between the physical model and the actual machining process, which improves the physical consistency and generalization of the model. Additionally, with the introduction of attention mechanism to the deep residual network, discriminant features can be extracted to distinguish wear values and working conditions. The experimental results show that by learning the physics, the physics-guided deep network can accurately predict the tool wear even under limited training sets and varied working conditions, and it outperforms the original data-driven models and the physical models.**

***Index Terms*—Deep learning, physics-guided data model, tool condition monitoring.**

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I. Introduction

As ONE of the important methods of modern manufacturing technology, computer numerical control (CNC) machining has been widely applied in aerospace, automotive, and medical industries due to its high productivity and high precision manufacturing capabilities [1], [2]. In smart CNC machining system, any changes of tool state may cause irreversible consequences and serious quality issues [1]. The development of online tool condition monitoring (TCM) system can not only avoid accidents, but also have a significant impact on the improvement of productivity and the reduction of cost [3], [4], [5]. At present, extensive studies have been carried out to predict the tool wear states, due to its importance and dominance in the TCM. Established tool wear state models are mainly divided into physics-based [6], data-driven [7], and hybrid models [8].

Physical based tool monitoring methods can be modeled through empirical formulas or by analyzing the tool wear mechanism. For example, Sipos [9] proposed an empirical fitting model based on exponential function and polynomial function representation, Zhu and Zhang [10] considered the tool wear properties in different stages, achieved a general wear model with adjustable coefficients. Archard [11] proposed the wear equation about wear rate and load and contact area. Later, Usui et al. [12] modified this equation for adhesive wear modeling by introducing the normal stress on the contact surface. For physical models, though these equations are convenient to establish and easy for application, they are built on simplified assumptions and often lead to large prediction error [28], [29].

For data-driven TCM approaches, currently, the commonly used methods mainly include deep neural networks [15], [16], [17], [18], [19], Gaussian process [20], [21], hidden Markov models [22], [23], random forests [24], etc. Wang et al. [15] proposed a temporal convolutional neural network with soft threshold and attention mechanism for remaining useful life estimation. Based on the residual network (Res-Net) [18], Zhao et al. [19] removed irrelevant data by introducing attention mechanism to improve the diagnostic performance. Kong et al. [20] proposed a tool wear assessment method based on the combination of kernel principal component analysis and Gaussian process regression (GPR). Lu et al. [25] proposed a

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class-imbalanced federated learning approach with biometric authentication for decentralized wind turbine fault diagnosis, which achieved good diagnostics with privacy protection. For data-driven approach, it need abundant labeled datasets to train the model, and the parameters of the model often difficult to be determined. The results also lack physical interpretation [26], [27].

For the hybrid models, they try to combine the benefits of two sides, and explore the knowledge both from the physics and the data, which often lead to better prediction accuracy [30]. The current research of hybrid models are mainly in three categories:

1) The hybrid model based on the combination of data model and physical model, which fuse the results to improve the model performance. Wang et al. [31] proposed a hybrid framework, which combines the results of physical based model and data driven model through particle filter algorithm to predict tool wear. Hanachi et al. [32] developed a regularized particle filter method to deal with the uncertainty of tool wear state.
2) The hybrid model based on physics-guided loss function, which constricts a regularized loss term through prior knowledge or physical model to improve the physical consistency and generalization ability of the data model. Wang and Zhang [33] established a tool wear prediction model with attention and monotone loss function, which captures degradation features through monotone loss function. Wang et al. [34] evaluated the physical inconsistency of model predictions by adding loss functions to the network. Howland and Ellis et al. [35] proposed a model based on integrated physics as a prior probability density function of the GPR model for the monitoring of turbomachine rotor blades.
3) The hybrid model based on pretraining with physics, which improves the model training and consistency with constrained model initialization. The use of simulation data generated by physical information to train data models has attracted attention in many engineering applications [36], [37], [38].

In the lake temperature prediction, Jia et al. [39] proposed a physics-guided recurrent neural network model (PGRNN). The model can combine the knowledge encoded in the physics-based model. This allowed few real observations to be used to train the PGRNN model, while still achieved high prediction accuracy.

The current hybrid model does not completely solve the respective defects of physical model and data model. For example, the studies [31], [32] were only loosely connected by fusing the independently predicted results from both sides. They can only make a limited improvement on the prediction results. When the prediction effect of the data model or physical model is very poor, it is also unable to accurately predict the tool failure. The approaches [33], [34] restricted the prediction of the data model through prior knowledge. Although its prediction results have been improved to some extent, it only uses the prior knowledge of the irreversibility of tool wear, and its physical information has not been fully utilized. Additionally, the training of the model need a large number of labeled datasets, and the model interpretability and generalization were poor [40], [41]. In order to solve the problems of the above models, training data models through simulation information generated by physical models has attracted more and more attention of the industry.

In this study, a physics-guided deep learning model is developed to solve the limitations of existing methods in generalization and prediction accuracy. First, the labeled dataset guided by the physical model is constructed through the physical model to solve the problem of insufficient samples of the training set in the data model training. Then pretrain the data model through the simulated dataset, and fine tune the model with a small number of real samples. Finally, the trained model is used to predict the tool state by integrating the process information with the physics-guided model. Compared with the current hybrid models, the physics implemented in this model is the mechanics of machining process rather than the general prior knowledge, so it can utilize the physics in-depth in the model construction. Moreover, the model can accurately predict the tool wear under different working conditions with only a small number of real samples for transfer learning, so the model has better generalization than the general hybrid model. It is noted that the authors' previous studies [5], [10], [43] have different ideas and approaches with this study. The study [10], [43] investigated the mechanisms of cutting process, and developed physical models of tool wear [10] and cutting force [43]. Then the physical models' applications to TCM were explored, the results of which form the basis of physics implemented in the deep network developed for this study. Zhu and Zhang [5] proposed a cyber-physical production system (CPPS) framework for machine tool monitoring, based on Big Data analytics. The study focused on the CPPS framework and monitoring system structure development. The ideas and approaches of these studies are very different. The main contributions to this study are as follows:

1) By learning the physics (mechanism) of the process, this study constructs a physics-guided training of the deep network that could solve the common problem of insufficient labeled samples and improve the physical consistency during the model training.
2) This study constructs a novel deep network structure, which realizes signal denoising and feature extraction synchronously by adding attention mechanism to deep residual network, so that the extracted features can distinguish wear values and working conditions.
3) This approach improves the generalization of the TCM model under varied working conditions.

II. Physics-Guided Data Model Structure

Based on the above analysis, this study develops a physics-guided data model to predict the tool states through tight modeling of both side information. The idea of model construction is shown in Fig. 1. First, it generates tool wear labels through the developed tool wear model (physical model) to solve the problem of insufficient labeled samples. Then, the force signal generated by force model is input to the constructed deep neural network for pretraining, and the model is then fine-tuned through real data. Finally, the milling force signal collected by the sensor is input into the trained network to predict the tool wear state.

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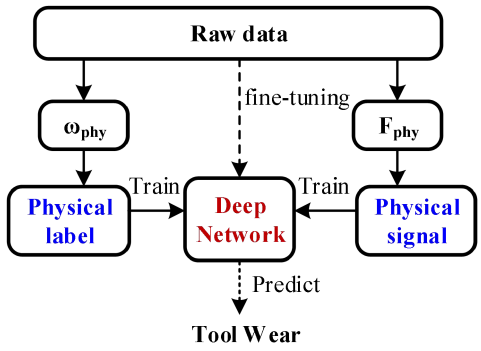


Fig. 1. Tool wear monitoring model framework.

The above process can be expressed by (1)–(4), in which (1) and (2) represent the generation of synthetic dataset through physical model ω_{phy}, F_{phy}, and (3) and (4) represent the pretraining process of data model $\theta_1$and the fine-tuning process of model parameters, respectively

$$L_{phy} = \omega_{phy}(L_{raw}) \tag{1}$$

$$D_{phy} = F_{phy}(D_{raw}) \tag{2}$$

$$\theta_1 = \theta_0 - \nabla_\theta \sum_{x_j, y_j \in D_{phy}, L_{phy}} \|F(x_j) - y_j\|^2_{\text{phy}} \tag{3}$$

$$\hat{\theta} = \theta_1 - \nabla_\theta \sum_{x_j, y_j \in D_{raw}, L_{raw}} \|F(x_j) - y_j\|^2_{\text{raw}}. \tag{4}$$

In the equations, ω_{phy} and F_{phy} represent tool wear model and force model, $F(x_j)$ and y_j are model output and target output, andθ_0,θ_1, $\hat{\theta}$ are the initial, pretraining, and after fine-tuning model parameters, respectively, ∇_θ represents the gradient of parameter θ. The detail structure of the physical-guided data model is shown in Fig. 2, and its components are discussed in the following sections.

A. Physics of Machining Process

The physics (mechanism) of tool wear and cutting force are most important in modeling the tool conditions of the machining process. The tool wear is inevitably generated due to excess friction and heat generated in the tool surface during the machining process. The full tool wear evolution can generally be divided into three stages: initial wear stage, stable wear stage, and accelerated wear stage. To model this process, Sipos [9] developed an empirical model with exponential and polynomial functions:

$$\omega(t) = \lambda t \exp(\alpha + \beta t + \gamma t^2) \tag{5}$$

where α, β, γ, and λ represent constant terms, ω represents the wear width of the flank, and t represents the milling time. Considering that in the actual tool wear process, tool wear is the result of the joint action of various wear mechanisms, and the tool wear rate is different in different wear stages. Zhu et al. [10] established a generic flank wear model with adjustable parameters by analyzing the wear rate in different stages

$$\omega(t) = A\text{In}(Bt + 1) + Ct^D \tag{6}$$

where A, B, C, and D represent the tool wear coefficient. The selection of tool wear model coefficient is obtained through calculation of experimental data. Compared with the model proposed by Sipos [9], the generic model (6) models the early and late stage of tool wear by considering the wear characteristics of tools at different wear stages, so it is more in line with the real tool wear law. Therefore, this study adopts the tool flank wear model established with (6).

Milling is one of the most representative machining processes due to its machining diversity and high quality. It is applied in this study. The cutting force of milling (milling force) is most important and sensitive in determining the tool conditions. The milling force of can be modeled as [42]

$$dF_i(\phi_j) = K_{ic} h(\phi_j) dz + K_{ie} dz (i = t, r, a) \tag{7}$$

where dF_i represents the cutting force element, K_{ic} is cutting force coefficient, K_{ie} is boundary force coefficient, t, r, and a are tangential, radial, and axial components, respectively, $h(\phi_j)$ is the instantaneous cutting thickness, and dz represents the length of the element, $K_{ic}h(\phi_j)dz$ is shear force, and $K_{ie}dz$ is tool tip friction force.

In precision milling, with consideration of the size effect, Li et al. [43] developed a generic instantaneous cutting thickness model considering cycloid trajectory and tool runout

$$h(\phi_j) = c\left(1+(-1)^j \tfrac{2R_0}{\pi R}\sin\gamma\right)\cdot\sin\phi_j - \tfrac{1}{\pi R}c^2\cdot\sin\phi_j\cos\phi_j + \tfrac{1}{2r}c^2\cdot\cos^2\phi_j - 2(-1)^j R_0\cos\gamma \tag{8}$$

where ϕ_j is the instantaneous immersion angle, c is the feed rate per tooth, R is the tool radius, and R_0 and γ represent tool runout length and angle, respectively. The runout parameters can be determined through calendar or heuristic algorithms, and are obtained through calendar in this article. The force coefficients K_{ic}, K_{ie} are evaluated by the recursive least square method [44].

B. Data-Driven Model for Tool Wear Monitoring

In the milling process, the wear of milling cutter will change the force signals collected by the sensor. Due to the sensitivity of the force signals to tool wear state, the high-dimensional nonlinear mapping between force signal and tool wear value can be established by data-driven model to predict tool wear value. The information flow of the data-driven model for predicting tool wear is shown in Fig. 3, and the details of the model structure are shown in the middle column of Fig. 2. The data-driven model are divided into two parts. The residual network with attention mechanism is used to extract the tool wear features and the bidirectional long short-term memory network (Bi-LSTM) is used to predict the wear value of the tool.

For feature extraction, while obtaining enough information to predict the different wear states of the tool under different working conditions, it is necessary to reduce the redundant information contained in the extracted features, thereby improving the prediction accuracy of pattern recognition. In this article, a

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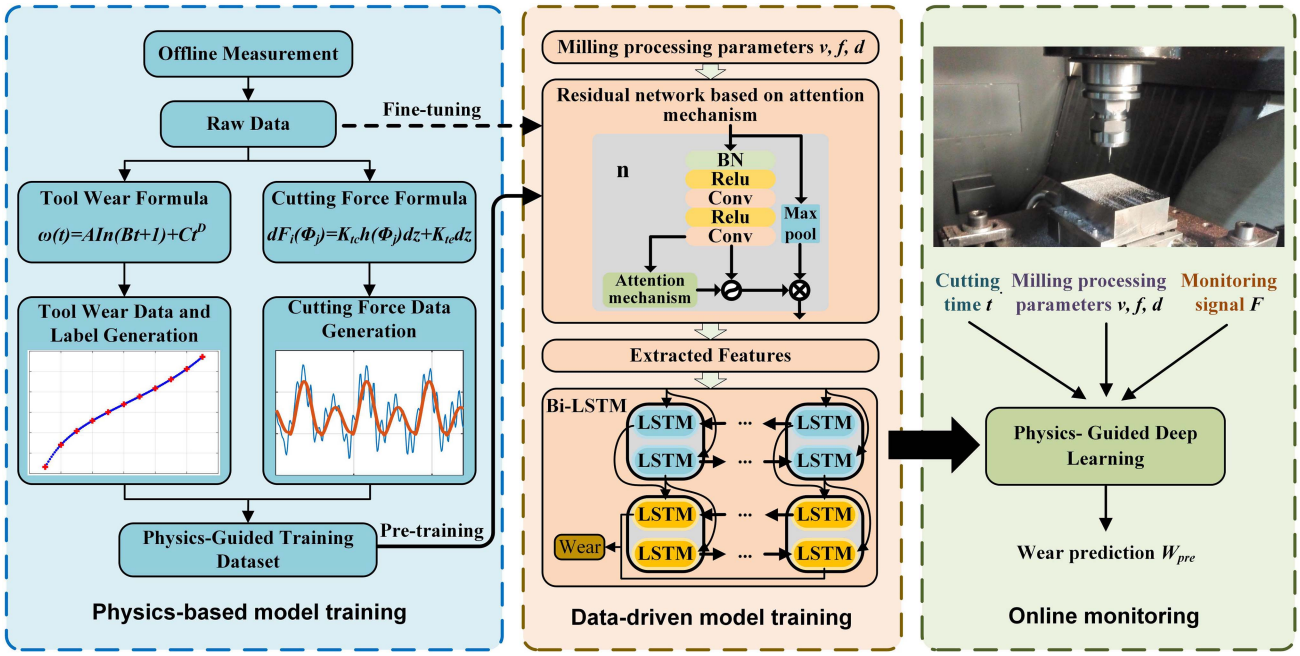


Fig. 2. Structure of the physics-guided data model for tool wear monitoring.

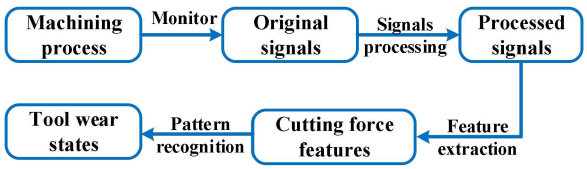


Fig. 3. Data-driven tool wear monitoring.

residual attention network is used to extract the characteristics of milling force signal. The network introduces a threshold of attention mechanism to the residual network, that is, a soft threshold f_s is inserted into the ResNet structure as a nonlinear transformation layer. The feature extraction can be expressed as

$$z_n = h(x_n) + f_s[G(x_n, w_n), t(G(x_n, w_n))] \tag{9}$$

$$x_{n+1} = \text{Relu}(z_n) \tag{10}$$

where x_n and x_{n+1} are the input and output of the nth residual unit, z_n represents the intermediate output of nth residual unit, G is the residual function, t and f_s are threshold functions, h represents the identity mapping function, and the Relu is an activation function. It is noted that the introduced attention mechanism acts the role of signal denoising in the model. The threshold f_s is automatically determined through the training process by using the gradient descent algorithm, which avoids the problem of manual experience determination. At the same time, the semisoft threshold method avoids the problem of vibration when the hard threshold method processes the signal, and also avoids the information loss problem of the soft threshold method.

The extracted milling force features $\{x_n\}$ are then input to Bi-LSTM network to predict the wear state. Compared with LSTM, Bi-LSTM can comprehensively consider historical information and future information, so as to improve the ability of tool wear prediction. Compared with RNN, Bi-LSTM introduces gating mechanism to adjust the path of information flow, and solves the problem of gradient disappearance in the process of RNN training. Bi-LSTM is a network composed of two LSTMs with the same structure. The information flow of the two LSTM networks is opposite. Taking the forward LSTM as an example, the information processing process is as follows:

$$f_n = \sigma(W_f h_{n-1} + U_f x_n + b_f) \tag{11}$$

$$i_n = \sigma(W_i h_{n-1} + U_i x_n + b_i) \tag{12}$$

$$a_n = \tanh(W_a h_{n-1} + U_a x_n + b_a) \tag{13}$$

$$c_n = c_{n-1} \odot f_n + i_n \odot a_n \tag{14}$$

$$o_n = \sigma(W_o h_{n-1} + U_o x_n + b_o) \tag{15}$$

$$h_n = o_n \odot \tanh(c_n) \tag{16}$$

where W and U represent the weight matrix of hidden state and input state, and b is the bias. The subscripts f, i, a, and o represent the variables belongs to forgetting gate, input gate, input gate, and output gate, respectively. "$\odot$" represents the Hadamard product, which multiplies corresponding elements in the matrices. Since the output of Bi-LSTM is the connection of the hidden states of the forward LSTM and the backward LSTM, there is: $h_n = [\vec{h}_n, \overleftarrow{h}_n]$. The optimization process of selecting weights and biases is carried out through the Adam optimizer, and its learning rate is set to 0.1.

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TABLE I
PSEUDOCODE OF PHYSICS-GUIDED DEEP LEARNING MODEL

Input: Synthetic datasets D_S; Real datasets D_r; Sensory signal S.
Output: Tool wear predicted y.

Initialization of the network weight with ω_0 and bias b_0 .

Pre-training process of the data model F using synthetic datasets D_S:

Evaluate $\nabla_{\omega,b} L(D_s)$ using D_s in

$$L(D_s) = \sum_{x^{(j)},y^{(j)} \in D_s} \left\| F(x^{(j)}) - y^{(j)} \right\|^2$$

Compute parameters with gradient descent:

$$\omega_1 = \omega_0 - \alpha \nabla_{\omega} L(D_s)$$

$$b_1 = b_0 - \alpha \nabla_{b} L(D_s)$$

Fine-tuning of data model F using real data sets D_r:

Evaluate $\nabla_{\omega,b} L(D_r)$ using D_r in

$$L(D_r) = \sum_{x^{(j)},y^{(j)} \in D_r} \left\| F(x^{(j)}) - y^{(j)} \right\|^2$$

Compute parameters with gradient descent:

$$\hat{\omega} = \omega_1 - \alpha \nabla_{\omega} L(D_r)$$

$$\hat{b} = b_1 - \alpha \nabla_{b} L(D_r)$$

Predict y from the trained data model F.

Return y

C. Physics-Guided Deep Learning Model

For the above physics-guided data model, the physics is represented as the tool wear and cutting force model. As shown in Fig. 1, the physical model first predicts the model parameters through the offline measured dataset, and then generates the simulation dataset through the physical model for the pretraining of the data model. It then applies the real dataset to fine-tune the data model to reduce the deviation caused by the inconsistency between the simulated dataset and the real dataset. For the tool wear monitoring stage, the milling force feature of tool wear is extracted through the residual network with an attention mechanism, and then input into a bidirectional long-term and short-term memory network to predict the tool wear value (see Fig. 2). The flow of the model algorithm is shown in Table I.

III. RESULTS AND DISCUSSION

A. Experimental Setup

In this study, milling experiments were carried out in a high-speed milling center to verify the effectiveness of the model. As shown in Fig. 4, it can be divided into experimental platforms and monitoring systems. In the experimental platform, the tools used in the experiments are two-flute cemented carbide tool, with diameter 0.5 mm and helix angle 30°. The workpiece is steel ($40CrNi_2Mo$) with a size of 30 mm × 60 mm. The Taguchi method is used to carry out milling experiment under nine working conditions (Table II). The dynamometer is used to obtain the milling force signals of each group, with sampling frequency 24000 Hz. The optical microscope is used to obtain the tool flank wear value offline. In the monitoring system, the upper computer is used to provide a computing environment for the model, so as to realize the training of the model and the prediction of the wear status. The lower computer is mainly used to establish the communication between the machine tool and the upper computer. The wear value calculated by the upper computer is transmitted to the lower computer through the PROFINET protocol. The lower computer first compiles the real type wear value into an IO signal, and then sends it to the IO module of the machine tool through IO Parallels. Finally, the machine tool optimizes the machining process in real time according to the wear value read by the IO module.

TABLE II
PARAMETER SETTING OF MILLING EXPERIMENT

Test	Spindle speed(n/rpm)	Cutting depth(a_p/μm)	Feed f_t (μm/tooth)	Number of measurements
1	18000	60	2	10
2	18000	80	4	10
3	18000	100	6	16
4	24000	80	6	16
5	30000	60	6	15
6	24000	60	4	9
7	24000	100	2	14
8	30000	80	2	10
9	30000	100	4	11

B. Data Generation With Physical Models

1) Expansion of Wear Labels: To overcome the limitations insufficient tool wear measurements in practice, the tool wear model is applied to generate labeled training data. To generate the labeled dataset, the first is to fit the coefficients in the model through a small number of labels measured offline to obtain the tool wear physical model under the corresponding working conditions, according to (6). Through the corresponding physical model to output the tool wear labels at different times, so as to expand the tool wear labels. The processing results of the wear label generated by the physical model are shown in Fig. 5. It is observed that with only 11 measured wear value, there are ten time more labeled samples generated by the wear equation.

2) Milling Force Signal Generation: The contradiction between a large number of labeled datasets and expensive costs in the data model. This article simulates the milling force through the physical model of milling force, and trains the data model through the simulated force, so as to predict the tool wear state and reduce the dependence of the data model on the real dataset. According to (7) and (8), the milling force is generated and applied to establish the relationship among the force, milling coefficient, and process parameters. The comparison diagram between the milling force simulated by the milling force model and the milling force monitored by the sensor is shown in Fig. 6.

It can be seen from the above Fig. 6 that the simulated milling force according to (7) is consistent with the real milling force both in time- and frequency-domains. For the real force, due to the measurement error and noise in the process, the real milling force has greater fluctuation. Although the real milling force has large noise fluctuation, its overall trend is in good agreement with the simulated milling force. Therefore, in this study, the milling force simulated by (7) replaces the real milling force to pretrain the deep network, which can improve the model training with much less experimental measurements.

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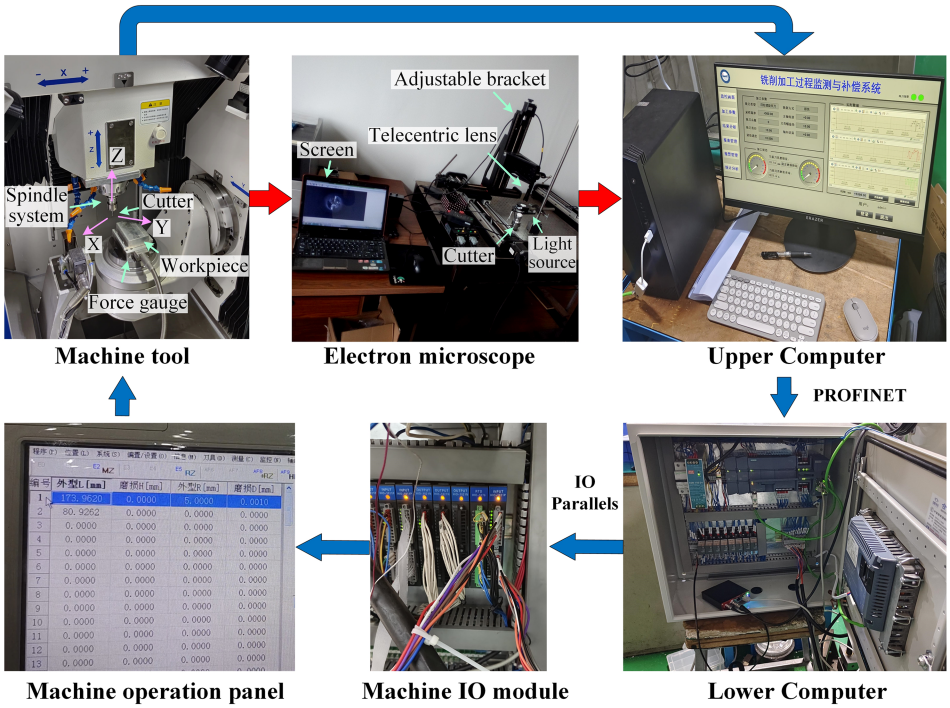


Fig. 4. Milling experimental platform and the approach implementation scheme.

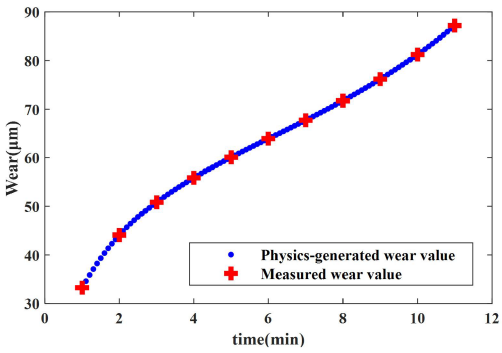


Fig. 5. Physical model to generate wear labels.

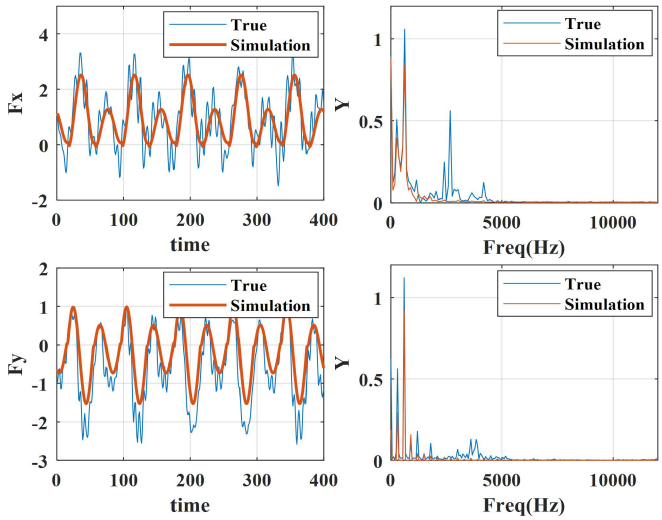


Fig. 6. Milling force simulation results.

C. Feature Extraction

The feature extraction process of the proposed method is shown in (9) and (10). Based on the strong representation capability, the ResNet is adapted to obtain the mapping relationship between milling force signal and feature, in which the attention mechanism is introduced for signal denoising and to reduce the mapping error. The traditional feature extraction method is to extract features from time domain, frequency domain, and time-frequency domain of milling force, including mean value, variance, maximum value, standard deviation, root mean square, average frequency, and wavelet energy spectral density. The features obtained by the above methods are visually represented by *t*-distributed random neighbor embedding algorithm (T-SNE), and evaluated by Davies Bouldin Index (DBI) [45]. The DBI measures the ratio of the sum of intraclass distances to interclass distances. The proposed attention mechanism-based threshold method is compared with the traditional threshold method (TTM), the traditional attention mechanism method (TAM), and the traditional feature extraction method (TFM). The results are shown in Fig. 7 and Table III.

TABLE III
PERFORMANCE OF VARIOUS MODELS

	Proposed method	TTM	TAM	TFM
DBI	0.77	1.53	2.26	3.98

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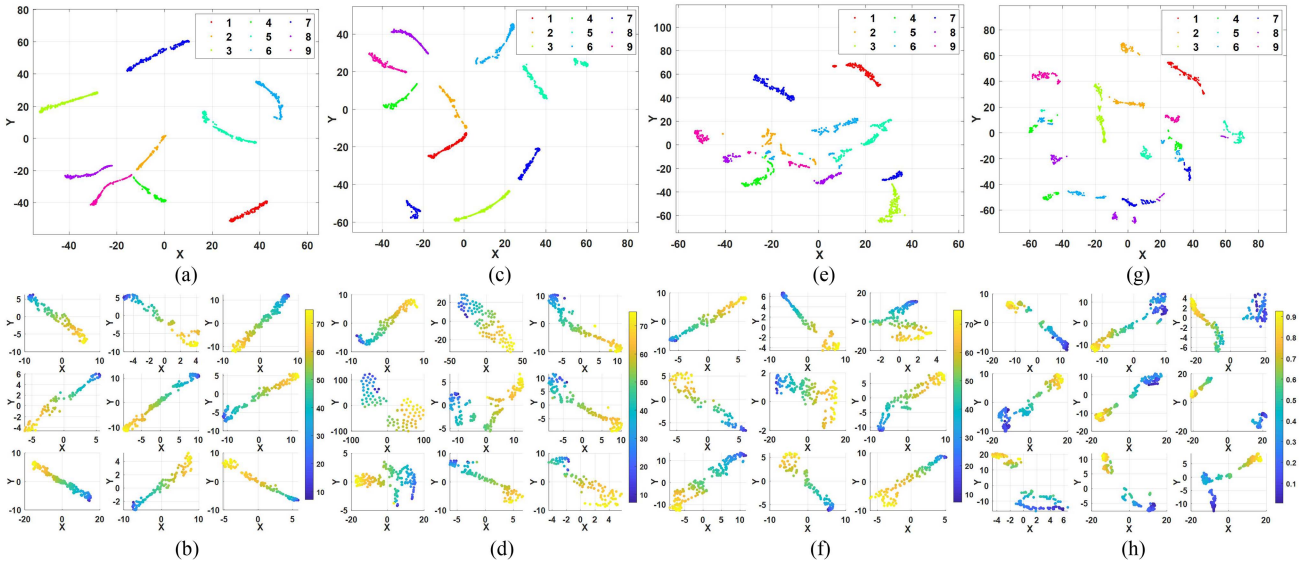


Fig. 7. T-SNE feature visualization. (a), (b) Proposed method. (c), (d) Method of simple threshold attention mechanism. (e), (f) Method of channel attention mechanism. (g), (h) Traditional feature extraction method.

It can be seen from Fig. 7(a), (c), (e), (g) that, compared with the features extracted by traditional threshold method, traditional attention mechanism method and traditional feature extraction method, the features extracted by deep network can better distinguish tool wear under different working conditions (Test 1–9). It is noted that with the T-SNE visualization, the scales of axis in Fig. 7 are not important and they are normalized to show the features' clustering and discriminative properties. In Fig. 7(b), (d), (f), and (h), the color distribution from blue to yellow on the right color bar shows the progression of tool from fresh to worn. It can be seen that the features extracted through the deep network can be used to distinguish not only the wear of tools, but also different cutting conditions. However, the traditional features extracted have a poor performance in distinguishing cutting conditions. Table III shows the DBI evaluation of the prediction effect of the above methods. It can be seen that the prediction effect of the proposed method is best among the four methods.

D. Model Performance Evaluation

In order to verify the proposed model performances, the prediction effects of the physics-guided data model (physics guided), physical model (physics based), and data-driven model (data driven) are compared by using MAPE, RMSE, and MFPE. MAPE index is used to indicate the deviation degree between the predicted results and the real results; RMSE index is used to represent the stability of model prediction; MFPE index is used to represent the maximum error value of model prediction results.

Table IV shows the prediction effects of different models. It can be seen that the prediction result of the physics-guided data model is better than that of physical model and data-driven model. This is because the prediction of the physical model is carried out in strict accordance with the established physical laws, and the prediction of the results is not related to the practical process. Under this circumstance, the accumulation of prediction errors of the physical model cannot be corrected in real time, and when the error is too large the prediction fails. For the data-driven model, the processing state provides feedback to the data-driven model online to avoid the problem of error accumulation. However, due to the lack of training samples, the data model cannot fully learn the correct relationship between model input and output, so the prediction of the model has large fluctuations. In view of the shortcomings of the current physical and data-driven models, a data-driven model based on physical information is proposed to monitor the wear state of cutting tools. It constructs the milling force dataset through the physical information of micromilling to expand the training samples and solve the problem of insufficient training samples of data model.

TABLE IV
PERFORMANCE OF VARIOUS MODELS

Model	MAPE	RMSE	MFPE
Physics-guided	**3.34%**	**2.22μm**	**7.63μm**
Data driven	6.93%	5.06μm	16.44μm
Physics-based	8.12%	5.40μm	20.03μm

The boldface represents the better results.

1) On the Effect of Physics on the Model Performance: The tool wear curves predicted by RMSE and five models under each machining condition are shown in Table V and Fig. 8.

Compare the proposed method (Proposed method) with physics-based model (Physics-based) and data driven model (Data driven). It can be seen that the performance of the proposed method is consistent and in general better than the physics-based and the data-driven models under different working conditions, with only an exception of Test 5 slightly higher than the physics-based method. It can be seen that the performance of the physics-guided data model is consistent and in general

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TABLE V
RMSE UNDER EACHMILLING TEST (μM)

Test	1	2	3	4	5	6	7	8	9	AVG
Proposed method	**1.73**	**1.79**	**1.91**	**2.77**	3.53	**2.37**	**2.47**	**1.63**	**1.75**	**2.22**
Data driven	4.47	4.10	7.24	5.74	6.97	3.94	5.25	5.65	2.24	5.06
Physics-based	4.83	3.83	8.75	11.66	**3.39**	2.70	4.82	5.41	3.22	5.40
Hybrid-model 1	3.69	2.30	4.84	6.53	3.62	2.95	4.48	1.82	2.77	3.66
Hybrid-model 2	5.51	3.07	4.69	5.76	3.87	4.34	4.85	2.40	4.84	4.37

The boldface represents the better results.

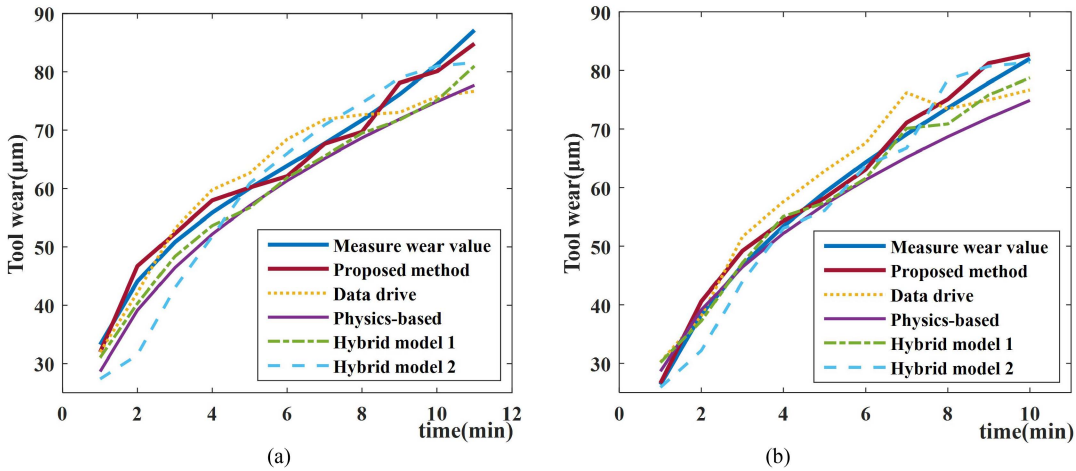


Fig. 8. Tool wear prediction results with: Proposed method, Data driven, Physics-based, Hybrid-model 1, Hybrid-model 2.

better than the physics-based and the data-driven models under different working conditions, with only an exception of Test 5 slightly higher than the physics-based method. The results indicate that the proposed model has a certain generalization ability under different working conditions. The improvement of generalization ability may stem from two aspects. From the perspective of training data, the dataset can be expanded by using the mechanics of micromilling to generate simulation data, so that the model can more fully learn the problems to be identified and learn more common knowledge. From the perspective of the model construction, this model extracts features by establishing a residual network based on the attention mechanism. The residual network can make the model avoid the disappearance of gradient in a very deep situation, and the deepening of the model can make it learn more abstract features, thus improving the representation capacity of the model. Moreover, it can be seen from Fig. 8 that due to the lack of real-time feedback of the processing state, the error of the physical model increases with the accumulation of error. Due to incomplete training, the tool wear curve of the data model fluctuates significantly.

Compare the proposed method with the classical physics-guided deep learning method, it can be seen that the proposed method has obvious advantages. For the Hybrid model 1 [32], it fuses the results of the physical model and the data model through the regularized particle filter technology. It takes the prediction results of the data model as the observation state, and the physical model calculates the conversion between the states. Because the pure data model is difficult to have a good prediction effect under small samples, and the first-order Taylor series of the physical model has a large error accumulation, the improvement of the results after particle filtering is very limited. For the Hybrid model 2 [34], Wang et al. considered the irreversibility of tool wear and used a loss function to constrain the relationship between current wear and previous wear. Although this constraint can improve the physical consistency of the data model to a certain extent, the selection of its physical loss weight and the size of the training sample will also have a great impact on the prediction results of the model.

TABLE VI
INFLUENCE OF TRAINING SET ON THE PERFORMANCE OF DATA MODEL

Model	MAPE	RMSE	MFPE
Simulation data	6.6%	4.71μm	13.84μm
Real data	6.93%	5.06μm	16.44μm
Simulation + real data	**3.34%**	**2.22μm**	**7.63μm**

The boldface represents the better results.

2) On the Effect of Datasets on the Model Performance: In order to verify the influence of the dataset on the performance of the tool wear state prediction data model, the experimental analysis and comparison of the model trained by the simulation dataset, the model trained by the real dataset, and the model jointly trained by the simulation dataset and the real dataset are shown in Table VI. It can be seen that the prediction results of the model trained by the simulation dataset and fine-tuned by the real data have better performance, and the prediction accuracy can be improved about 50%. This is because for the deep learning

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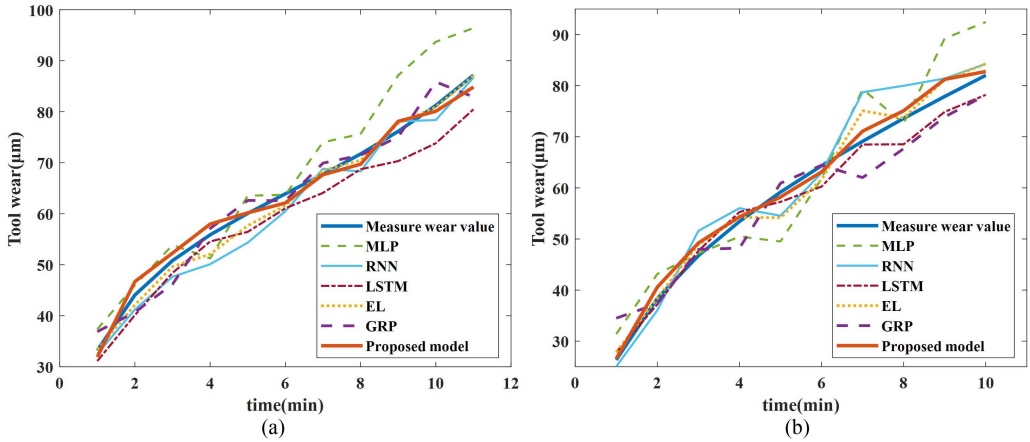


Fig. 9. Tool wear prediction results with: MLP, RNN, LSTM, EL, GRP, proposed model.

TABLE VII
RMSE UNDER EACH MILLING TEST (μM)

Test	1	2	3	4	5	6	7	8	9	AVG
MLP	6.60	7.07	7.43	5.46	7.14	6.53	5.25	6.00	6.19	6.41
RNN	3.32	4.60	5.49	3.07	5.30	2.58	5.73	3.29	5.01	4.26
LSTM	4.31	2.80	3.31	5.08	3.75	3.88	3.93	3.31	2.85	3.69
EL	1.79	3.02	4.57	2.49	4.15	2.82	4.26	3.33	2.80	3.25
GPR	3.04	4.64	6.28	7.98	6.19	5.61	4.93	7.29	2.91	5.43
Proposed model	1.73	1.79	1.91	2.77	3.53	2.37	2.47	1.63	1.75	2.22

model, it relies on a large training dataset to adjust the weight of each neuron and the value of bias term through error back propagation, so that the model can establish an accurate high-dimensional mapping relationship between the input and the output. The accuracy of model is determined by the size and varieties of the training set.

In practice, due to the diversity of machining parameters, it is impossible to obtain a large amount of labeled training data. The acquisition of tool wear data is costly and time-consuming. Therefore, when only the limited real dataset is used to train the model, it is insufficient and the predicted result deviate greatly from the actual value. Because the milling force formula is established under ideal conditions, the actual machining conditions are much more complex than the ideal conditions. Therefore, the prediction results of the model trained only by simulation data are not satisfactory. In this case, the developed approach can pretrain the model through the simulation data, and then fine tune the model through a small amount of real data. It not only solve the quantity of training data, but also makes up for the deviation of simulation data. This makes the data model have a better prediction accuracy in the case of less real data.

3) Comparison With the State-of-the-Art Methods: In order to verify the proposed method, the predicted results of the physics-guided data model are compared with those of MLP, RNN, LSTM, the ensemble learning model (EL) [46], and the GPR [47], as shown in Fig. 9 and Table VII. The parameter settings for the models are: The MLP model consists of three full connective layers, and the number of neurons is 100, 20, and 1, respectively. The structure of RNN model is ResNet18, and its convolution core size is set to 240 × 1 and 10 × 1. The channel is set to 128 and 256. The RNN model consists of two simple RNN layers and two complete connection layers. The hidden layer status dimension of RNN is set to 64. The LSTM model consists of two LSTM layers and two full connection layers. The status dimension of the hidden layer of LSTM is set to 64. For all the above model, the activation function is Relu, the loss function is MSE, and the optimizer is Adam. For the EL model, its heterogeneous ensemble used MLP, RNN, and LSTM individual models as base learners, and they were integrated using the ensemble hard voting strategy. In this strategy, each unique base model was trained on the same dataset, and the classification output of each base model was used as a "vote" to determine the final prediction. For the GPR model, its mean function was set to zero mean, the covariance function was set to squared exponential (SE) covariance, and the initial values of all covariance function hyperparameters were set to 1.

It can be observed in Table VII and Fig. 9 that the prediction results based on the proposed model are optimal. The MLP model, RNN model, LSTM model, EL model, and GPR model are all data-driven models that can theoretically fit any complex function with sufficient training data. However, their performance could be significantly decreased when the training data are insufficient. In contrast, the proposed method can guide the data model by learning the physics embedded in case of labeled training data insufficiency, thereby ensuring the model prediction accuracy and generalization. Therefore, compared with the aforementioned models, the proposed method has the best prediction performance.

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IV. Conclusion

In order to efficiently utilize both the machining mechanism and monitored data for TCM, this study constructs a physics-guided deep learning model to predict the tool wear state. The model performances are validated with experimental studies and compared with state-of-the-art methods. The following conclusions can be drawn:

1) The proposed physics-guided deep learning network can make up the shortcomings of the data-driven model by learning the physics of machining process, which improves the physical consistency and generalization of the data-driven model.
2) The network can be effectively initialized by pretraining through the simulation dataset constructed by the physical model, which alleviates the labeled-data insufficiency and training difficulty of the deep network.
3) The attention mechanism based deep residual network can achieve signal denoising and high-dimensional discriminant feature extraction synchronously, and contribute to establish an end-to-end tool wear condition monitoring model.

Further study will be carried on the extension of physical models to achieve tool wear compensation and process control, as well as on the validation of the models' performance on different machining types such as turning and drilling to improve the generalization of this approach.

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